Human Re-Identification for Crowd Management

SESSION 2022-2026

Abstract

Traditional crowd-counting systems tally detections rather than unique individuals, causing duplicate counts when people leave and re-enter monitored spaces. This project presents an identity-consistent Human Re-Identification framework for event and public-infrastructure crowd management. Using 25,000 camera-captured frames, the system integrates YOLOv8 person detection, ByteTrack short-term tracking, and a ResNet-50 body Re-ID model fine-tuned on project-specific data. Body-only matching is compared with a combined body–face fusion system, where facial cues supplement body features when available. A dynamic identity gallery stores learned body and face representations, then matches returning individuals using cosine similarity. In test-video evaluation, the combined system achieved proper re-identification , improving identity consistency under re-entry, pose variation, occlusion, and similar clothing while supporting reduced duplicate counting and reliable crowd statistics.

1. Introduction

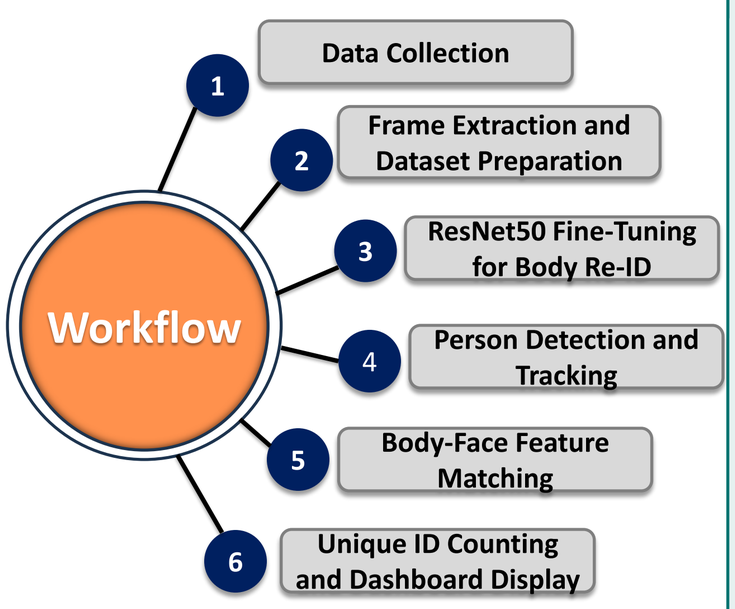
- ❑ Conventional counters tally appearances, not unique individuals, affecting facility decisions.
- ❑ Repeated counting of returning visitors distorts occupancy data and resource planning.
- ❑ Human Re-ID addresses this gap by distinguishing newcomers from returning persons.
- ❑ Pipeline: YOLOv8 → ByteTrack → ResNet-50 fine-tuned on 25,000 project-specific frames.
- ❑ Body features anchor identity; face cues reinforce matching when visible.

UN Sustainable Development Goals

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

11 SUSTAINABLE CITIES AND COMMUNITIES

2. Block Diagram, Tools and Technologies

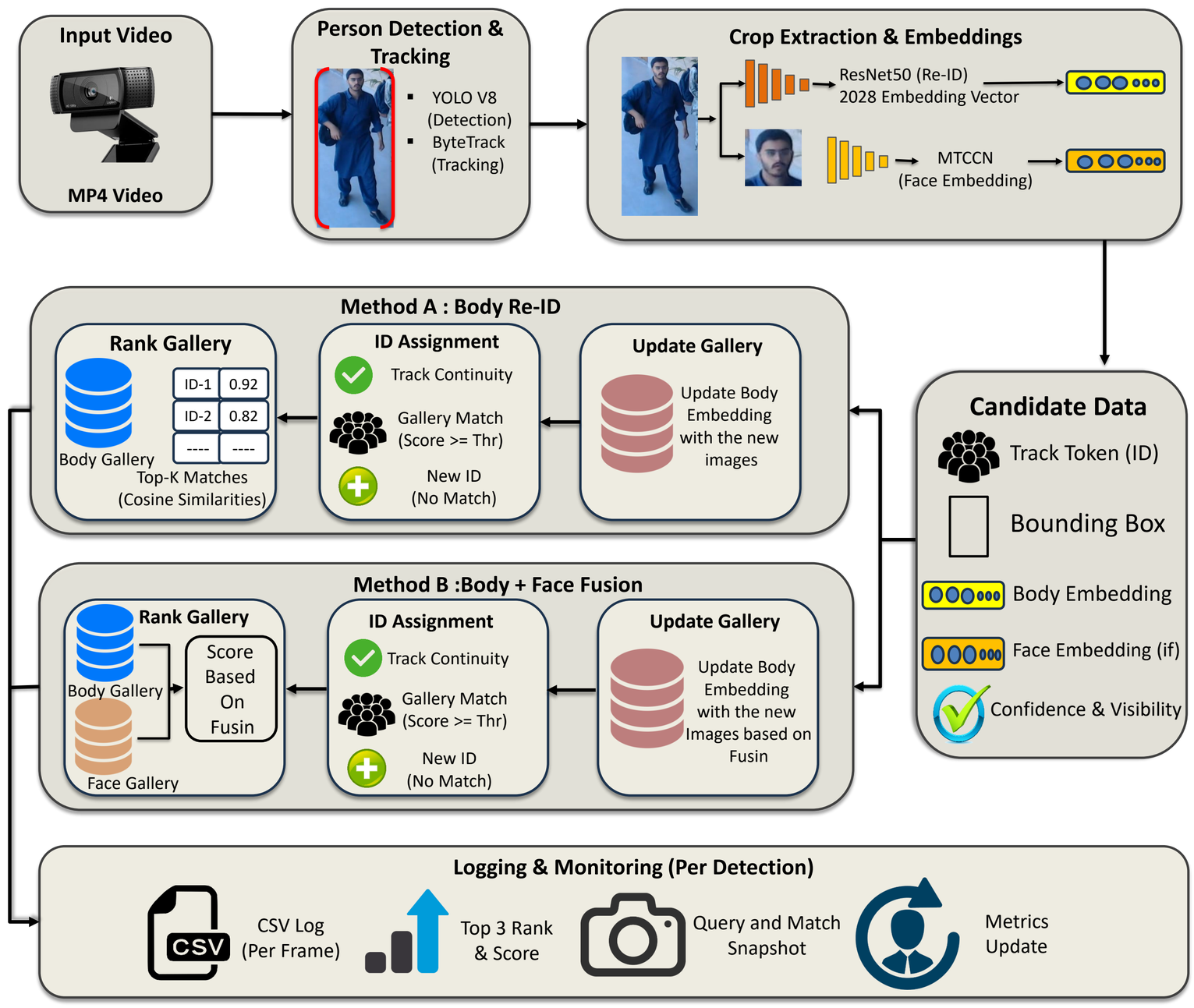


Tools and Technologies

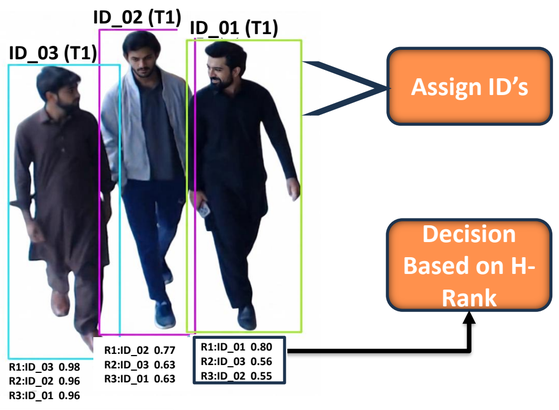
PyTorch NumPy YOLO

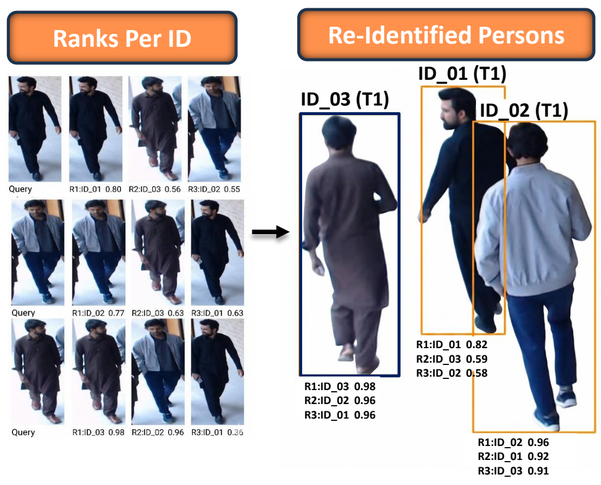
3. Design, Results, and Analysis

Design



Results & Discussion





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Limitations

- Fully-covered individuals prevent reliable feature extraction
- Severe occlusion causes temporary ID tracking loss
- Dense crowd crossing introduces ID swap errors
- Lighting variations require camera-specific recalibration
- Identical clothing reduces body feature discrimination

Conclusion

This project shows that crowd counting becomes more reliable when identity is preserved across repeated appearances. Instead of counting every detected person as a new entry, the system checks whether the same person has appeared before. YOLOv8 detects people, ByteTrack supports short-term tracking, and a fine-tuned ResNet-50 extracts body-based identity features from 25,000 project-specific camera frames. A dynamic identity gallery then matches new detections with stored identities using cosine similarity. Facial features are added only when visible, giving extra support when body appearance is affected by pose change, partial occlusion, re-entry, or similar clothing. Our system indicates that body–face fusion can support duplicate-count reduction and improve unique-person statistics in controlled crowd-management scenarios.

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